Restricting a normal quotient search to divisors of its target order

We use the previously reviewed finite-target Coxeter enumeration and the LINS normal-subgroup algorithm. Their completeness and the supplied finite group tables remain computational premises. This note justifies a further restriction when the desired quotient has a prescribed order. It then applies that restriction to the last square diagram in Table 3 of Fernandes–Leemans–Weiss, *An Exploration of Locally Spherical Regular Hypertopes*, fixed 16-page edition, physical page 14.

Lemma 1. *Let $N \lhd W$ have finite index m. Every subgroup H containing N has index dividing m. In particular, every normal overgroup of N is retained by the condition $[W : H] \mid m$.*

Proof. The index formula gives $m = [W : H][H : N]$. □

The search and its dependencies. The acquired search first supplies the required normal kernels of finite target epimorphisms. Its later operations descend through normal p-group sections and take intersections of already known normal subgroups. The finite-target list, module calculations and kernel construction are unchanged. We apply the following restrictions:

1. A finite epimorphic target Q is used only if $|Q| \mid m$.
2. A node H is expanded only if $[W : H] \mid m$.
3. A p-section below H is considered only if $p[W : H] \mid m$; a proposed elementary section is retained only if its resulting index $[W : K]$ divides m.
4. An intersection is retained only if its computed index divides m.

The ordinary bound remains m. No early termination is introduced. In particular, "divides m" is not replaced by "equals m" at an intermediate step.

To justify completeness, fix a desired N and follow the acquired algorithm's completeness argument inside the finite normal-subgroup interval $[N, W]$. Every epimorphic target used in this interval is a quotient of W/N, so its order divides m. Every intermediate subgroup in a descent contains N. Its index and each intervening elementary section therefore satisfy the tests above. If an intersection produces a subgroup containing N, both operands also contain N, so neither operand nor their intersection is removed. Lemma 1 thus preserves every dependency needed to obtain N.

There is also an implementation issue concerning the routine which decides whether a p-section is already obtainable by intersections. For any retained H, every normal supergroup of H has index dividing $[W : H]$, hence dividing m. Its complete supergroup interval is retained. Consequently the minimal-supergroup relations used by that routine have not changed. The same applies to the common-supergroup computation for an intersection. The existing correctness of these optimizations is therefore inherited on the retained interval. It is not inferred from small tests alone.

Exact implementation and controls. The program uses LINS's documented option hooks for branch cuts, p-quotients, proposed modules and intersections. The initial finite-target routine is the acquired Coxeter routine with one additional order test. The exact presentation is checked there; its involution choices include the identity and proper divisors of bond orders, because a quotient on the way to the desired group may collapse a generator or a bond. All final marked quotients are tested separately for exact orders. The program requires a positive target within the installed table bound, and requires the search bound to equal that same target.

For $A_3, B_3, C_2^3, \widetilde{A}_2, H_3$, the controls compare the restricted search with the full default search. There are 26 target choices and 132 retained normal kernels. Each comparison checks the actual two-way subgroup containment, not merely its index. The H_3 cases exercise the nonabelian finite targets. The elementary abelian cases retain many distinct kernels of the same index. A deliberately incorrect restriction to intermediate indices equal to the final target loses kernels, confirming why normal ancestors must be preserved. Invalid target values and a mismatched search bound must be rejected before a search can be credited.

The square with labels $3, 5, 3, 5$. Put

$$W = \langle s_1, s_2, s_3, s_4 \mid s_i^2 = 1,\ (s_1s_2)^3 = (s_2s_3)^5 = (s_3s_4)^3 = (s_4s_1)^5 = 1,\ (s_1s_3)^2 = (s_2s_4)^2 = 1\rangle.$$

Every finite regular hypertope with this diagram gives a quotient of this marked presentation. Its four distinguished generators must satisfy

$$H_I \cap H_J = H_{I\cap J}, \qquad H_I = \langle t_i : i \in I\rangle.$$

This necessary condition applies whether a finite-diagram residue is required to be a full spherical Coxeter geometry or a projective quotient.

For target $m = 9720$, the retained normal-kernel indices are

$$1, \quad 2, \quad 60, \quad 120, \quad 4860, \quad 9720,$$

with one kernel at each index. The unique kernel of index 9720 gives the 27-point marked quotient bound in the literal certificate. It has the exact required pair orders and four maximal parabolics of order 120. Nevertheless it fails the intersection property. With indices starting at 1, the literal closure gives

$$|H_{\{1,2,3\}} \cap H_{\{1,2,4\}}| = 12, \qquad |H_{\{1,2\}}| = 6.$$

There is a shorter obstruction in the same tuple. For each omitted j, form $z_j = (\prod_{i\neq j} t_i)^5$, with the remaining indices increasing. All four words give the same nonidentity involution z. Thus z belongs to every maximal parabolic, whereas repeated application of the intersection property would make their total intersection trivial. The equality and nonidentity of these four words are verified directly on the 27 points; full group closure is unnecessary for this obstruction.

Theorem 2. *There is no regular hypertope of square type* $(3, 5, 3, 5)$ *whose type-preserving group has order 9720, conditional on the stated normal quotient enumeration and its retained classification premises.*

Proof. Any such hypertope supplies an index-9720 normal kernel of W. The complete restricted enumeration contains exactly one. Its marked quotient fails the necessary intersection property as shown above. Relabelling a diagram gives another marking of this same fixed presentation and does not evade its normal-kernel enumeration. □

Attribution, reuse and scope. The target order and the displayed group $3^4 \rtimes A_5 \times C_2$ come from the second $(3, 5, 3, 5)$ entry of the source table. The conclusion excludes the order itself and hence does not depend on interpreting an unspecified action in that group name. It is a correction of a published finite claim, not a solution of an open conjecture.

The normal-subgroup algorithm and its finite group tables are due to prior work; the installed documentation attributes the underlying algorithm to Firth's thesis. The finite-target Coxeter traversal, composed module evaluation, complete marked-intersection test and independent literal permutation checker are acquired from earlier reviewed components. The contribution of this note is the explicit justification and use of the divisor restriction, together with its exact source application. Divisibility of finite indices is elementary; no historical priority is claimed for that principle.

The earlier complete search through order 3840 is retained separately. It contains 25 normal kernels, including eleven of index 3840, and gives one valid broad marked hypertope of order 1920 with maximal parabolics of order 60. Its raw finite result and its local interpretation are not silently incorporated into the theorem here. The general bound 9720 previously timed out; that failed computation is preserved and provides no completeness premise. The present theorem concerns the exact target order, not the entire interval below it, and makes no minimum claim.

The certificate checker reruns the bounded restricted computation, compares the ordered marking with the recorded one by an actual marked isomorphism, checks the GAP-to-JSON bridge, and applies the independent integer-permutation closure to all 256 intersections. It separately checks the four displayed central words and malformed input controls. The proof, source table and dependency bytes are retained for distinct review. There is no whole-manuscript credit, new Lean verification, or compression measurement in this component. Other claims and examples in the source remain attached to their own recovery obligations.